

APPLIANCE GUARD To Prevent Damage

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This circuit monitors the AC current flowing through an electrical appliance and automatically turns the appliance off when the current goes beyond maximum or minimum threshold levels. If the appliance is drawing very high or very low current than the specified rated current, this indicates that there is a fault with the appliance. This guard is extremely useful where motors are used in the appliance or an equipment.

Circuit and working

The circuit diagram of the appliance guard is shown in Fig. 1. It is built around microcontroller (MCU) ATmega8A (IC1), op-amp LM358 (IC2), 5V voltage regulator 7805 (IC3), 16x2 LCD (LCD1), two transistors BC547 (T1) and 2N2222 (T2), step-down transformer (X1), 12V relay (RL1) and a few other components.

The circuit measures AC current flowing through the appliance/equipment by continuously monitoring the current value, which is also displayed on the LCD. AC current drawn by the appliance is measured using current transformer CT1, which has many turns of wire wound on a circular core. AC current carrying the conductor (say, phase, line or live wire) is passed through

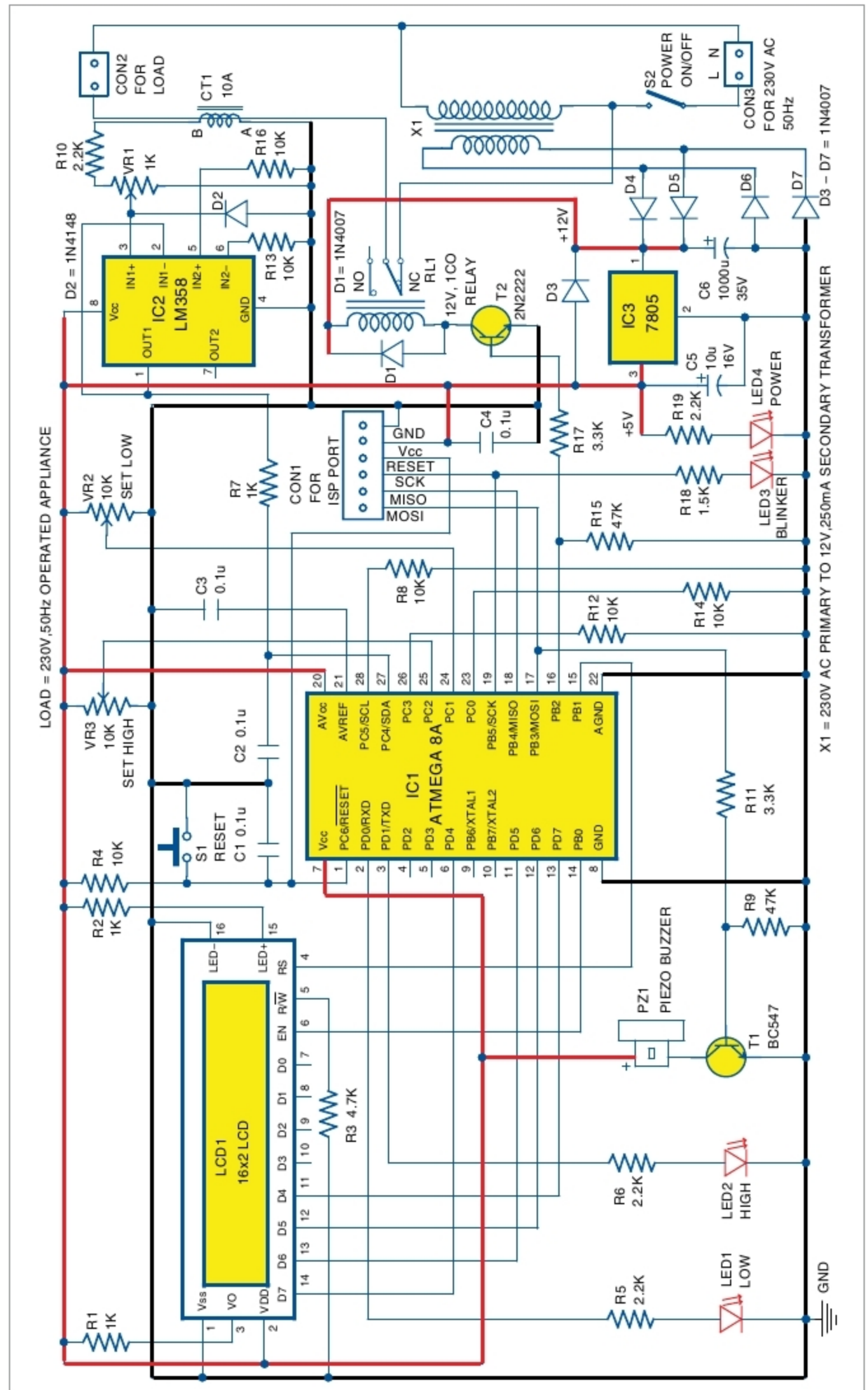


Fig. 1: Circuit diagram of appliance guard